# A superquadratic separation between sensitivity and block sensitivity

Rewritten for readability from the OpenAI manuscript of the same title

September 25, 2026

**Abstract**

We disprove the quadratic strengthening of the Sensitivity Conjecture by constructing nonconstant total Boolean functions whose block sensitivity grows faster than any constant multiple of sensitivity squared. In fact, for some fixed $\alpha > 2$, composed examples (Corollary 4.3) have unbounded block sensitivity and satisfy $\mathrm{bs}(f) \geq s(f)^{\alpha}$.

## 1 Introduction

**Definitions.** For a positive integer $n$ write $[n] = \{1, \ldots, n\}$. For $x \in \{0,1\}^n$ and $B \subseteq [n]$, let $x^B$ be $x$ with exactly the coordinates in $B$ flipped, and write $x^e = x^{\{e\}}$. (The source writes $x^a$; we call a single coordinate $e$ because $a(i,j)$ will denote edge labels.) In particular $0^B$ is the indicator string of $B$. For a total Boolean function $f : \{0,1\}^n \to \{0,1\}$,

$$s(f,x) = \big|\{e \in [n] : f(x^e) \neq f(x)\}\big|, \qquad s(f) = \max_x s(f,x).$$

A nonempty $B \subseteq [n]$ with $f(x^B) \neq f(x)$ is a *sensitive block* of $f$ at $x$. The block sensitivity $\mathrm{bs}(f,x)$ is the largest number of pairwise disjoint sensitive blocks at $x$, and $\mathrm{bs}(f) = \max_x \mathrm{bs}(f,x)$.

**Theorem 1.1.** *For every real $C > 0$, there are a positive integer $n$ and a nonconstant total Boolean function $f : \{0,1\}^n \to \{0,1\}$ such that*

$$\mathrm{bs}(f) > C\, s(f)^2.$$

*More precisely, for every integer $d \geq 1$ the construction below gives a nonconstant total Boolean function $f$ with*

$$\frac{\mathrm{bs}(f)}{s(f)^2} \geq \frac{2^d}{4(d+2)^2}.$$

Section 4 also proves Corollary 4.3: a fixed $\alpha > 2$ and functions $F_m$ with $\mathrm{bs}(F_m, 0) \geq s(F_m)^{\alpha}$ and $\mathrm{bs}(F_m, 0) \to \infty$. Since $\mathrm{bs}(F_m) \geq \mathrm{bs}(F_m, 0)$, this is the second claim of the abstract.

**Background.** The Sensitivity Conjecture asked whether block sensitivity is bounded by a polynomial in sensitivity. It goes back to Nisan, who introduced block sensitivity and recorded Rubinstein's quadratic separation [Nis89, Section 2]; Nisan and Szegedy suggested $\mathrm{bs}(f) \leq s(f)^2$ [NS94, Section 4]. Rubinstein's construction [Rub95] showed that a polynomial bound needs degree at least two; Virza obtained $\mathrm{bs}(f) = \frac{1}{2}s(f)^2 + \frac{1}{2}s(f)$ [Vir11] and Ambainis and Sun $\mathrm{bs}(f) = \frac{2}{3}s(f)^2 - \frac{1}{3}s(f)$ [AS11, Theorem 1]. Huang's degree bound with Tal's refinement [Tal13] gives $\mathrm{bs}(f) \leq s(f)^4$ [Hua19,

Theorem 1.5], improved by Wellens to $\mathrm{bs}(f) \le \sqrt{2/3}\, s(f)^4 + 1$ [Wel22, Corollary 4.2]. Theorem 1.1 rules out every bound $\mathrm{bs}(f) \le Cs(f)^2$.

The architecture (tournament-indexed rows, one selected blocking coordinate in each outgoing row, random labels) adapts Meiburg's spectral-sensitivity construction [Mei26, Sections 3.1 and 5.2–5.5][1], here with nested predicates and a different forbidden-label condition (Lemma 2.1). Spectral sensitivity $\lambda(f)$ [ABDK+21] satisfies $\lambda(f) \le s(f)$, so a lower bound on $\mathrm{bs}(f)/\lambda(f)^2$ does not give Theorem 1.1; the estimates below are proved directly. The final OR follows Ambainis and Sun [AS11, Section 4, Lemma 1]. Ambainis and Prūsis noted that a seed with $\mathrm{bs} > s^2$ yields a superquadratic power separation by iteration [AP14, Section 1]; composition is treated systematically by Tal [Tal13].

**Proof at a glance.** For levels $\ell = 0, \dots, d$ we build predicates $P_{\ell,1} \ge P_{\ell,2} \ge \cdots$ (pointwise). A level-$\ell$ input is a $k \times r$ array of level-$(\ell-1)$ inputs ("children"), with $k = 2M^2 + 1$ rows. Each $P_{\ell,q}$ is an OR of $k$ clauses, one per row. Clause $i$ requires the strongest child predicate to accept every child of row $i$ (*targets*) and a weaker child predicate to reject one labelled child in each of $M^2$ other rows (*gates*); a tournament on the rows decides which rows, and the index $q$ decides which weaker predicate. Each level multiplies the number of disjoint sensitive blocks at 0 by $k > 2M^2$, but the sensitivity by only about $M$; so $\mathrm{bs}/s^2$ gains a factor $> 2$ per level.

1. Lemma 2.1: a labelling of tournament edges avoiding a certain 16-vertex pattern (union bound). Used only in step C2 of Proposition 3.1.
2. Lemma 2.2: sensitivity bounds at level 0. Starts the induction in Lemma 4.1.
3. Lemma 2.3: $Mk^\ell$ disjoint sensitive blocks at 0 on level $\ell$. Used in the proof of Theorem 1.1.
4. Proposition 3.1 (the core): recurrences for $S_{\ell,b}$ (number of flips taking a predicate from $b$ to $1-b$) and $J_{\ell,b}$ (number of flips taking two predicates from $b$ to $1-b$ simultaneously). The $J$'s are needed because repairing a failed gate usually forces *two* child predicates to change, which is much rarer than one.
5. Lemma 4.1: for $M \ge 9^{d+1}$, $S_{\ell,b} \le 2(d+2)M^{\ell+b}$ (from items 2 and 4).
6. Theorem 1.1: an OR of $M$ copies of $P_{d,1}$ balances $S_{d,0}$ against $S_{d,1}$ (from items 3 and 5).
7. Lemma 4.2 and Corollary 4.3: composing the $d = 9$ function with itself.

## 2 The labelled tournament and nested predicates

Fix integers $d \ge 1$ and $M \ge 3$ ($M$ is chosen in terms of $d$ in Section 4), and put

$$L = d + 2, \qquad k = 2M^2 + 1, \qquad r = \lceil \sqrt{M} \rceil, \qquad t = 16.$$

A tournament orients every pair of distinct vertices: exactly one of $i \to j$, $j \to i$ holds. Fix a tournament on $[k]$ in which every vertex has outdegree $M^2$; for example, place the $2M^2 + 1$ vertices on a cycle and direct each vertex to the next $M^2$ vertices.

**Lemma 2.1.** *The edges can be labelled by numbers $a(i,j) \in [r]$ so that there is no set $I \subseteq [k]$ of size $t$ admitting choices $m_j \in [r]$, $j \in I$, with*

$$a(i,j) = m_j \qquad \text{for every edge } i \to j \text{ within } I.$$

[1] In the attribution accompanying this construction, Meiburg credits GPT-5.6-Sol with the gated-tournament ideas and describes his own contributions as exposition, size and parameter optimization, and checking the faithfulness of the Lean formalization.

*Proof.* Choose all labels independently and uniformly from $[r]$. Fix $I$ with $|I| = 16$ and $(m_j)_{j\in I}$. Every pair of vertices of $I$ carries exactly one edge, so there are $\binom{16}{2} = 120$ edges within $I$. The stated equalities prescribe the labels of these 120 distinct edges, so they all hold with probability $r^{-120}$. There are at most $\binom{k}{16} r^{16} \le k^{16} r^{16}$ choices of $(I, (m_j))$, so the probability that some choice satisfies all its equalities is at most

$$k^{16} r^{16} r^{-120} = k^{16} r^{-104} \le (3M^2)^{16} (\sqrt{M})^{-104} = 3^{16} M^{32-52} = 3^{16} M^{-20} \le 3^{-4} < 1,$$

using $k = 2M^2 + 1 \le 3M^2$, $r \ge \sqrt{M}$, and $M \ge 3$. Hence a suitable labelling exists. □

Fix such a labelling. For $0 \le \ell \le d$ set

$$H_\ell = d + 1 - \ell, \qquad N_\ell = LM(kr)^\ell.$$

We define predicates $P_{\ell,q} : \{0,1\}^{N_\ell} \to \{0,1\}$, $1 \le q \le H_\ell$, satisfying

**(N)** *Nesting.* $P_{\ell,1}(x) \ge P_{\ell,2}(x) \ge \cdots \ge P_{\ell,H_\ell}(x)$ for every $x$.

Thus if $P_{\ell,q}(x) = 0$ then every predicate of larger index is 0 at $x$, and if $P_{\ell,q}(x) = 1$ then every predicate of smaller index is 1 at $x$. Nesting concerns the index $q$, not the input bits.

## 2.1 Initial predicates

Partition $[LM]$ into $M$ blocks $E_1, \ldots, E_M$ of size $L$ and let $z_j = 0^{E_j}$. With dist the Hamming distance, set

$$D(x) = \min_{j\in[M]} \operatorname{dist}(x, z_j), \qquad P_{0,q}(x) = \mathbf{1}\{D(x) \le d + 1 - q\} \quad (1 \le q \le d+1).$$

So $P_{0,q}$ indicates the union of the Hamming balls of radius $d+1-q \in \{0, \ldots, d\}$ around the centres $z_j$. (N) holds because the radius decreases with $q$. Distinct centres are at distance $2L = 2d + 4$.

## 2.2 Recursive clauses

Let $1 \le \ell \le d$. Put $H = H_\ell$ and $Q = H + 1 = H_{\ell-1}$, and abbreviate

$$p_\gamma = P_{\ell-1,\gamma} \qquad (1 \le \gamma \le Q)$$

(our shorthand); by (N) at level $\ell - 1$, $p_1 \ge \cdots \ge p_Q$, so $p_Q$ is the strongest. Split $x \in \{0,1\}^{N_\ell}$ into $kr$ disjoint children $x_{j,c} \in \{0,1\}^{N_{\ell-1}}$, with row $j \in [k]$ and position $c \in [r]$; this uses $N_\ell = krN_{\ell-1}$. For $i \in [k]$ and $q \in [H]$, *clause $i$ at index $q$* asserts

$$\begin{aligned} p_Q(x_{i,c}) &= 1 \quad \text{for all } c \in [r] && (\textit{target conditions}),\\ p_{Q-q}(x_{j,a(i,j)}) &= 0 \quad \text{for all } j \text{ with } i \to j && (\textit{gate conditions}). \end{aligned} \tag{1}$$

Define $P_{\ell,q}(x) = 1$ if clause $i$ at index $q$ holds for some $i$, and 0 otherwise. Indices are valid: $1 \le Q - q \le H < Q$. Two facts are used throughout.

**(DC)** *Distinct children.* For fixed $i$, the $r + M^2$ conditions of clause $i$ sit on distinct children, the same ones for every $q$: targets occupy row $i$, and the gates lie in distinct rows $j \ne i$ (there is one edge from $i$ to $j$). A single coordinate of $x$ lies in one child, so flipping it changes the truth value of at most one condition of clause $i$. (One child may occur in many clauses; we handle this by union bounds.)

**(Mon)** *Monotonicity in $q$.* If $q < q'$ and clause $i$ holds at index $q'$, then it holds at index $q$: the targets are the same, and $p_{Q-q'}(y) = 0$ implies $p_{Q-q}(y) = 0$ by (N), because $Q - q > Q - q'$.

(Mon) gives $P_{\ell,q} \ge P_{\ell,q'}$ for $q < q'$, which is (N) at level $\ell$.

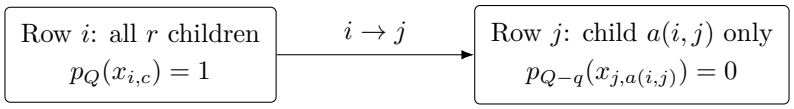



Figure 1: Clause $i$ at index $q$: $r$ targets in row $i$, and one gate in each of the $M^2$ rows $j$ with $i \to j$.

### 2.3 Sensitivity profiles and the initial bounds

For $b \in \{0,1\}$ define

$$S_{\ell,b} = \max_{\substack{1 \le q \le H_\ell,\ x \in \{0,1\}^{N_\ell} \\ P_{\ell,q}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = 1 - b\}\big|,$$

$$J_{\ell,b} = \max_{\substack{1 \le q < q' \le H_\ell,\ x \in \{0,1\}^{N_\ell} \\ P_{\ell,q}(x) = P_{\ell,q'}(x) = b}} \big|\{e \in [N_\ell] : P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1 - b\}\big|.$$

So $S_{\ell,b}$ bounds the number of coordinates whose flip moves one predicate from $b$ to $1-b$, and $J_{\ell,b}$ the number whose flip moves two distinct predicates from $b$ to $1-b$ at once. An empty maximum is zero; in particular $J_{d,0} = J_{d,1} = 0$, since $H_d = 1$.

**Lemma 2.2.** *The initial sensitivity profiles satisfy*

$$S_{0,0} \le L, \qquad S_{0,1} \le LM, \qquad J_{0,0} = J_{0,1} = 0. \tag{2}$$

*Proof.* $S_{0,0}$. Fix $q$, put $R = d + 1 - q \in \{0, \dots, d\}$, and let $x$ satisfy $D(x) > R$. Suppose $D(x^e) \le R$, so $\mathrm{dist}(x^e, z_j) \le R$ for some $j$; then $\mathrm{dist}(x, z_j) \le R + 1 \le d + 1$. At most one centre lies within distance $d + 1$ of $x$, since two would be within $2d + 2 < 2L$ of each other. So $j$ does not depend on $e$. Also $\mathrm{dist}(x, z_j) \ge D(x) \ge R + 1$, hence $\mathrm{dist}(x, z_j) = R + 1$, and $e$ must be one of the $R + 1$ coordinates where $x$ and $z_j$ differ. This gives at most $R + 1 \le d + 1 \le L$ flips.

$S_{0,1}$. There are only $LM$ coordinates.

$J_{0,b}$. If $x, y$ differ in one coordinate, the triangle inequality gives $|D(x) - D(y)| \le 1$. Let $q < q'$, with radii $R = d + 1 - q$ and $R' = d + 1 - q'$, so $R \ge R' + 1$. If both predicates are 0 at $x$ and 1 at $x^e$, then $D(x) \ge R + 1 \ge R' + 2$ and $D(x^e) \le R'$. If both are 1 at $x$ and 0 at $x^e$, then $D(x) \le R'$ and $D(x^e) \ge R + 1 \ge R' + 2$. Either way $D$ changes by at least two, which is impossible. □

### 2.4 Disjoint blocks at zero

**Lemma 2.3.** *At every level $0 \le \ell \le d$, all predicates are zero at the all-zero input. There are $Mk^\ell$ pairwise disjoint nonempty blocks of coordinates, each of size $Lr^\ell$, such that flipping any one block at zero makes every predicate at that level equal to one.*

*Proof.* Induction on $\ell$. *Level 0:* $\mathrm{dist}(0, z_j) = L$ for all $j$, so $D(0) = L > d$ and all predicates vanish at 0. The $M$ blocks $E_j$ work, since $0^{E_j} = z_j$ has $D = 0$.

*Level $\ell \ge 1$:* at the all-zero input every child is all-zero, so $p_Q = 0$ on every child by induction; every target, hence every clause, fails, and all $P_{\ell,q}(0) = 0$. Let $B_1, \dots, B_{Mk^{\ell-1}}$ be the level-$(\ell-1)$ blocks. For each row $i$ and each $h$, let $B_{i,h}$ be the union of the copies of $B_h$ in the $r$ children of row $i$. After flipping $B_{i,h}$ at 0:

- each child of row $i$ equals $0^{B_h}$, so $p_Q = 1$ there by induction: all targets of clause $i$ hold;
- children in other rows are still all-zero, so every $p_\gamma$ is 0 on them: all gates of clause $i$ hold, for every $q$.

Thus every $P_{\ell,q}$ equals one. The $B_{i,h}$ are pairwise disjoint (different rows use different children; within a row the $B_h$ are disjoint in each child). There are $k \cdot Mk^{\ell-1} = Mk^{\ell}$ of them, each of size $r \cdot Lr^{\ell-1} = Lr^{\ell}$. □

## 3 Sensitivity recurrences

**Proposition 3.1.** *For $1 \le \ell \le d$, write $S'_b = S_{\ell-1,b}$ and $J'_b = J_{\ell-1,b}$. Then*

$$\begin{aligned} S_{\ell,1} &\le M^2 S'_0 + rS'_1, & J_{\ell,1} &\le M^2 J'_0 + rS'_1, \\ S_{\ell,0} &\le tS'_0 + S'_1 + 3J'_1, & J_{\ell,0} &\le tS'_0 + 3J'_1. \end{aligned} \tag{3}$$

*Plan.* A flip changes a parent predicate only by breaking or completing a clause, and by (DC) it does so through exactly one condition on one child. We count, child by child, the flips that reverse that condition, using $S'_b$ when one child predicate must change and $J'_b$ when two must change. Parts A–B treat inputs of value 1 (one fixed satisfied clause); Parts C–D treat inputs of value 0, where the task is to bound the number of clauses one flip away from being satisfied.

*Proof.* Fix $\ell$ and use $H$, $Q$, $p_\gamma$ as in Section 2. In Parts B and D we write $\gamma = Q - q$, $\gamma' = Q - q'$ (the source's $g, g'$; renamed because $g$ is a function in Lemma 4.2). A "flip" always means flipping one coordinate $e$ of the level-$\ell$ input; $e$ lies in exactly one child. If $H = 1$ the two $J$-inequalities hold trivially (empty maximum), so in Parts B and D assume $H \ge 2$.

**Three tools.** By the definitions of $S'_b$ and $J'_b$, for every child $y$ and $b \in \{0, 1\}$:

**(S′)** for each $\gamma \in [Q]$, at most $S'_b$ coordinates of $y$ have a flip changing $p_\gamma(y)$ from $b$ to $1 - b$;

**(J′)** for each $1 \le \gamma' < \gamma \le Q$, at most $J'_b$ coordinates of $y$ have a flip changing both $p_{\gamma'}(y)$ and $p_\gamma(y)$ from $b$ to $1 - b$.

**(SF)** *Single-flip principle.* Let $w, w'$ differ in one coordinate $e$, and suppose clause $i$ (at a fixed index) fails at $w$ and holds at $w'$. Then exactly one condition of clause $i$ fails at $w$, and it sits on the child containing $e$. Reason: some condition fails at $w$, none fails at $w'$, and by (DC) only the condition on the child containing $e$ can differ between $w$ and $w'$.

**Part A: $S_{\ell,1} \le M^2 S'_0 + rS'_1$.** Fix $q$ and $x$ with $P_{\ell,q}(x) = 1$, and fix one clause $i$ that holds at $x$. If $P_{\ell,q}(x^e) = 0$, clause $i$ fails at $x^e$, so by (SF) (with $w = x^e$, $w' = x$) the coordinate $e$ lies in one of the $r + M^2$ children of clause $i$ and reverses the condition there. On a target child this means $p_Q$ goes $1 \to 0$: at most $S'_1$ choices of $e$ by (S′). On a gate child it means $p_{Q-q}$ goes $0 \to 1$: at most $S'_0$ choices. Summing over $r$ target and $M^2$ gate children gives the bound.

**Part B: $J_{\ell,1} \le M^2 J'_0 + rS'_1$.** Fix $q < q'$ and $x$ with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 1$; then $1 \le \gamma' < \gamma < Q$. Fix a clause $i$ that holds at $x$ at index $q'$; by (Mon) it also holds at index $q$. Let $e$ satisfy $P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 0$. Since $P_{\ell,q}(x^e) = 0$, clause $i$ fails at $x^e$ at index $q$. By (SF) at index $q$, $e$ lies in a child $y$ of clause $i$ whose index-$q$ condition holds before and fails after the flip.

- *$y$ is a target child.* Then $p_Q(y)$ goes $1 \to 0$: at most $S'_1$ such $e$ per target child, $rS'_1$ in all.
- *$y$ is a gate child.* Before the flip, $p_{\gamma'}(y) = 0$ (the index-$q'$ gate holds) and $p_\gamma(y) = 0$ (the index-$q$ gate holds). After the flip $p_\gamma(y) = 1$ (the index-$q$ gate fails), hence $p_{\gamma'}(y) = 1$ by (N). Both $p_{\gamma'}$ and $p_\gamma$ go $0 \to 1$, so by (J′) there are at most $J'_0$ such $e$ per gate child, $M^2 J'_0$ in all.

**Part C:** $S_{\ell,0} \le tS_0' + S_1' + 3J_1'$**.** Fix $q$ and $x$ with $P_{\ell,q}(x) = 0$; every clause fails at $x$. Put $\gamma = Q - q < Q$.

*C1 (candidate lists).* Let $\mathcal{T}$ be the set of $i \in [k]$ whose clause fails exactly one target and no gate at $x$, and $\mathcal{G}$ the set of $i$ whose clause fails exactly one gate and no target. Both depend only on $x$ and $q$. If $P_{\ell,q}(x^e) = 1$, some clause $i$ holds at $x^e$; by (SF), $i \in \mathcal{T} \cup \mathcal{G}$, and $e$ lies in the child of the unique failed condition of clause $i$ and makes that condition true. For $i \in \mathcal{T} \cup \mathcal{G}$ let $\rho_i$ be the number of coordinates of that child whose flip makes that condition true. By the union bound, the number of $e$ with $P_{\ell,q}(x^e) = 1$ is at most $\sum_{i\in\mathcal{T}} \rho_i + \sum_{i\in\mathcal{G}} \rho_i$.

*C2 ($|\mathcal{T}| \le t-1$ and $\rho_j \le S_0'$ for $j \in \mathcal{T}$).* For $j \in \mathcal{T}$ let $m_j \in [r]$ be the position of its failed target: $p_Q(x_{j,m_j}) = 0$ and $p_Q(x_{j,c}) = 1$ for $c \ne m_j$. Let $i, j \in \mathcal{T}$ with $i \to j$. Clause $i$ fails no gate, so $p_\gamma(x_{j,a(i,j)}) = 0$, and by (N) (as $\gamma < Q$) $p_Q(x_{j,a(i,j)}) = 0$. The only position of row $j$ where $p_Q$ vanishes is $m_j$, so $a(i,j) = m_j$. If $|\mathcal{T}| \ge t$, any $t$-subset $I \subseteq \mathcal{T}$ with the numbers $(m_j)_{j\in I}$ would contradict Lemma 2.1. Hence $|\mathcal{T}| \le t-1$. Repairing clause $j \in \mathcal{T}$ means changing $p_Q(x_{j,m_j})$ from 0 to 1, so $\rho_j \le S_0'$ by (S′), and $\sum_{j\in\mathcal{T}} \rho_j \le tS_0'$.

*C3 ($|\mathcal{G}| \le 3$).* For $j \in \mathcal{G}$ all targets hold:

$$p_Q(x_{j,c}) = 1 \ \text{ and hence, by (N), } \ p_\gamma(x_{j,c}) = 1 \qquad \text{for all } c \in [r]. \tag{$\star$}$$

Let $i, j \in \mathcal{G}$ with $i \to j$. By ($\star$), $p_\gamma(x_{j,a(i,j)}) = 1$, so the gate of clause $i$ on edge $i \to j$ fails. Clause $i$ has exactly one failed gate, so $i$ has at most one out-neighbour in $\mathcal{G}$. Write $w = |\mathcal{G}|$ (the source's $m$). The tournament restricted to $\mathcal{G}$ has $w(w-1)/2$ edges and at most $w$ (one per tail), so $w(w-1)/2 \le w$, i.e. $w \le 3$. Moreover at most one vertex of $\mathcal{G}$ has no out-neighbour in $\mathcal{G}$, since of any two vertices one points to the other.

*C4 (counting $\mathcal{G}$).* Let $i \in \mathcal{G}$.

(a) *$i$ has an out-neighbour $j \in \mathcal{G}$.* By C3 its unique failed gate sits on $y = x_{j,a(i,j)}$, where $p_\gamma(y) = p_Q(y) = 1$ by ($\star$). Repairing the gate makes $p_\gamma(y) = 0$, and then $p_Q(y) = 0$ by (N). So two distinct child predicates ($\gamma < Q$) go $1 \to 0$, and $\rho_i \le J_1'$ by (J′). (That this flip breaks a target of clause $j$ is irrelevant: we only bound $\rho_i$ from above.)

(b) *$i$ has no out-neighbour in $\mathcal{G}$* (at most one such $i$). Repairing its failed gate changes $p_\gamma$ from 1 to 0 on one child, so $\rho_i \le S_1'$ by (S′).

Since $|\mathcal{G}| \le 3$, $\sum_{i\in\mathcal{G}} \rho_i \le S_1' + 3J_1'$. With C1 and C2 this proves Part C.

**Part D:** $J_{\ell,0} \le tS_0' + 3J_1'$**.** Fix $q < q'$ and $x$ with $P_{\ell,q}(x) = P_{\ell,q'}(x) = 0$; again $1 \le \gamma' < \gamma < Q$. Let $\mathcal{T}, \mathcal{G}$ be the lists of Part C for $x$ and the *smaller* index $q$; so $|\mathcal{T}| \le t-1$ and $|\mathcal{G}| \le 3$. Let $e$ satisfy $P_{\ell,q}(x^e) = P_{\ell,q'}(x^e) = 1$, and choose a clause $i$ that holds at $x^e$ at index $q'$ (the choice may depend on $e$; the lists do not). By (Mon) clause $i$ holds at $x^e$ at index $q$, and it fails at $x$ at index $q$. By (SF) at index $q$, $i \in \mathcal{T} \cup \mathcal{G}$ and $e$ lies in the child $y$ of its unique index-$q$ failed condition.

- $i \in \mathcal{T}$. Then $e$ changes $p_Q(x_{i,m_i})$ from 0 to 1; as in C2 there are at most $S_0'$ such $e$ per $i$, at most $tS_0'$ in all.
- $i \in \mathcal{G}$. Then $y$ is the child of the failed index-$q$ gate, so $p_\gamma(y) = 1$ before the flip, and $p_{\gamma'}(y) = 1$ by (N). After the flip clause $i$ holds at index $q'$, so its gate on $y$ gives $p_{\gamma'}(y) = 0$, and then $p_\gamma(y) = 0$ by (N). Both $p_{\gamma'}$ and $p_\gamma$ go $1 \to 0$, so by (J′) there are at most $J_1'$ such $e$ per $i \in \mathcal{G}$, at most $3J_1'$ in all.

The second case never used ($\star$): the second changing predicate is $p_{\gamma'}$, supplied by the index $q'$, instead of $p_Q$. So it covers the vertex of case C4(b) too, and no $S_1'$ term appears. (The witness $i$ also has exactly one failed condition at index $q'$ at $x$, by (SF) at index $q'$; members of $\mathcal{T} \cup \mathcal{G}$ with more failures at index $q'$ are never witnesses and are merely over-counted.)

Every bound holds for all admissible $q$ (or $q < q'$) and $x$; taking maxima gives (3). □

## 4 Quantitative bounds and separation

**Lemma 4.1.** *Suppose that $d \geq 1$ and $M \geq 9^{d+1}$. For every $0 \leq \ell \leq d$ and $b \in \{0,1\}$, the predicates of the construction satisfy*

$$S_{\ell,b} \leq 2LM^{\ell+b}.$$

*Plan.* The dominant terms $M^2 S'_0$ and $S'_1$ in (3) suggest dividing $S_{\ell,b}$ and $J_{\ell,b}$ by $M^{\ell+b}$. This gives a two-term linear recurrence in which the joint quantity is fed only through a small coefficient $\epsilon$ but can triple at each level. The hypothesis $M \geq 9^{d+1}$ makes $\epsilon \leq 2/3^{d+1}$, which beats $3^d$. Used in the proof of Theorem 1.1.

*Proof.* Define

$$u_\ell = \max_{b\in\{0,1\}} \frac{S_{\ell,b}}{M^{\ell+b}}, \qquad v_\ell = \max_{b\in\{0,1\}} \frac{J_{\ell,b}}{M^{\ell+b}}, \qquad \epsilon = \frac{\max(r,t)}{M}.$$

*Step 1 (base).* By (2), $S_{0,0}/M^0 \leq L$ and $S_{0,1}/M \leq L$, so $u_0 \leq L$; and $v_0 = 0$.

*Step 2 (normalised recurrences).* Let $1 \leq \ell \leq d$. By definition $S'_b \leq M^{\ell-1+b}u_{\ell-1}$ and $J'_b \leq M^{\ell-1+b}v_{\ell-1}$. Dividing (3) by $M^{\ell+1}$ (case $b=1$) or $M^\ell$ (case $b=0$):

$$\begin{aligned}
\frac{S_{\ell,1}}{M^{\ell+1}} &\leq \frac{S'_0}{M^{\ell-1}} + \frac{r}{M}\cdot\frac{S'_1}{M^\ell} \leq \Big(1+\frac{r}{M}\Big)u_{\ell-1},\\
\frac{S_{\ell,0}}{M^{\ell}} &\leq \frac{t}{M}\cdot\frac{S'_0}{M^{\ell-1}} + \frac{S'_1}{M^\ell} + \frac{3J'_1}{M^\ell} \leq \Big(1+\frac{t}{M}\Big)u_{\ell-1} + 3v_{\ell-1},\\
\frac{J_{\ell,1}}{M^{\ell+1}} &\leq \frac{J'_0}{M^{\ell-1}} + \frac{r}{M}\cdot\frac{S'_1}{M^\ell} \leq \frac{r}{M}u_{\ell-1} + v_{\ell-1},\\
\frac{J_{\ell,0}}{M^{\ell}} &\leq \frac{t}{M}\cdot\frac{S'_0}{M^{\ell-1}} + \frac{3J'_1}{M^\ell} \leq \frac{t}{M}u_{\ell-1} + 3v_{\ell-1}.
\end{aligned}$$

Taking maxima over $b$, and using $v_{\ell-1} \leq 3v_{\ell-1}$,

$$u_\ell \leq (1+\epsilon)u_{\ell-1} + 3v_{\ell-1}, \qquad v_\ell \leq \epsilon u_{\ell-1} + 3v_{\ell-1}. \tag{4}$$

*Step 3 (size of $\epsilon$).* Since $M \geq 9^{d+1} \geq 81$, we have $r = \lceil\sqrt{M}\rceil \leq \sqrt{M}+1 \leq 2\sqrt{M}$ and $t = 16 \leq 2\sqrt{M}$. With $\sqrt{M} \geq 3^{d+1}$,

$$\epsilon \leq \frac{2}{\sqrt{M}} \leq \frac{2}{3^{d+1}}. \tag{5}$$

*Step 4 (comparison sequences).* Let $U_0 = L$, $V_0 = 0$ and

$$U_\ell = (1+\epsilon)U_{\ell-1} + 3V_{\ell-1}, \qquad V_\ell = \epsilon U_{\ell-1} + 3V_{\ell-1}.$$

All coefficients are nonnegative, so Step 1 and (4) give $u_\ell \leq U_\ell$, $v_\ell \leq V_\ell$ by induction on $\ell$.

*Step 5.* We show by induction that for $0 \leq \ell \leq d$,

$$U_\ell \leq 2L, \qquad V_\ell \leq \epsilon L(3^\ell - 1). \tag{6}$$

Both hold at $\ell = 0$. Assume them for all indices below some $1 \leq \ell \leq d$. First,

$$V_\ell = \epsilon U_{\ell-1} + 3V_{\ell-1} \leq 2\epsilon L + 3\epsilon L(3^{\ell-1}-1) = \epsilon L(3^\ell - 1).$$

Next, $U_{j+1} - U_j = \epsilon U_j + 3V_j$; summing over $j < \ell$ and using $\sum_{j=0}^{\ell-1}(3^j - 1) = \frac{1}{2}(3^\ell - 1) - \ell$,

$$U_\ell = L + \sum_{j=0}^{\ell-1}(\epsilon U_j + 3V_j) \leq L + 2\epsilon L\ell + 3\epsilon L\Big[\frac{3^\ell - 1}{2} - \ell\Big] = L + \epsilon L\Big[\frac{3}{2}(3^\ell - 1) - \ell\Big].$$

By (5) and $\ell \le d$,

$$\epsilon\Big[\frac{3}{2}(3^\ell - 1) - \ell\Big] \le \frac{2}{3^{d+1}} \cdot \frac{3^{\ell+1} - 3 - 2\ell}{2} = \frac{3^{\ell+1} - 3 - 2\ell}{3^{d+1}} < 1,$$

so $U_\ell < 2L$. This completes the induction. Hence $S_{\ell,b} \le M^{\ell+b} u_\ell \le M^{\ell+b} U_\ell \le 2LM^{\ell+b}$. □

*Proof of Theorem 1.1.* Fix an integer $d \ge 1$ and an integer $M \ge 9^{d+1}$ (so $M \ge 3$), and a labelling from Lemma 2.1. Following Ambainis and Sun [AS11, Section 4, Lemma 1], let $f$ be the OR of $M$ copies of $P_{d,1}$ on disjoint sets of variables. Each copy has $LM(kr)^d$ variables, so $f$ is a total Boolean function on

$$n = LM^2(kr)^d \tag{7}$$

bits.

*Step 1 ($f(0) = 0$, and block sensitivity).* By Lemma 2.3 every copy is 0 at its zero input, so $f(0) = 0$. Each copy has $Mk^d$ pairwise disjoint blocks whose flip at zero makes that copy, hence $f$, equal to 1. Over the $M$ disjoint copies this gives $M^2k^d$ pairwise disjoint sensitive blocks of $f$ at 0:

$$\mathrm{bs}(f, 0) \ge M^2k^d. \tag{8}$$

In particular $f$ is nonconstant, so $s(f) > 0$.

*Step 2 (sensitivity).* Let $f(x) = 0$. Then every copy is 0, and a flip changes $f$ only if it turns its own copy to 1; there are at most $S_{d,0}$ such flips per copy, so $s(f, x) \le MS_{d,0}$. Let $f(x) = 1$. If at least two copies accept, no single flip makes $f$ zero. If exactly one accepts, a sensitive flip must lie in that copy and turn it to 0, so $s(f, x) \le S_{d,1}$. By Lemma 4.1,

$$s(f) \le \max\{MS_{d,0}, S_{d,1}\} \le \max\{M \cdot 2LM^d,\ 2LM^{d+1}\} = 2LM^{d+1}. \tag{9}$$

*Step 3 (ratio).* By (9), (8), and $k/M^2 > 2$,

$$\frac{\mathrm{bs}(f)}{s(f)^2} \ge \frac{\mathrm{bs}(f, 0)}{s(f)^2} \ge \frac{M^2k^d}{4L^2M^{2d+2}} = \frac{(k/M^2)^d}{4L^2} \ge \frac{2^d}{4(d+2)^2}. \tag{10}$$

This proves the second statement. The right-hand side tends to infinity with $d$. Given $C > 0$, choose $d$ with $2^d/(4(d+2)^2) > C$, then $M$, the labelling and $f$ as above; since $s(f) > 0$, $\mathrm{bs}(f) > Cs(f)^2$. □

### 4.1 A fixed exponent by composition

Composing the $d = 9$ function with itself gives a power separation, by the composition principle studied by Tal [Tal13]. We prove the two facts needed. In the next lemma the source's dimensions $a, b$, block counts $p, q$ and blocks $C_i, D_j$ are renamed $n_1, n_2$, $p_1, p_2$, $X_i, Y_j$ to avoid clashes with earlier notation.

**Lemma 4.2.** *Let $f : \{0,1\}^{n_1} \to \{0,1\}$ and $g : \{0,1\}^{n_2} \to \{0,1\}$ be total Boolean functions. Their disjoint composition is the function on $n_1n_2$ bits defined by*

$$(f \circ g)(x_1, \dots, x_{n_1}) = f\big(g(x_1), \dots, g(x_{n_1})\big), \qquad x_i \in \{0,1\}^{n_2}.$$

*Then*

$$s(f \circ g) \le s(f)s(g).$$

*If $g(0) = 0$, then $(f \circ g)(0) = f(0)$ and*

$$\mathrm{bs}(f \circ g, 0) \ge \mathrm{bs}(f, 0)\,\mathrm{bs}(g, 0).$$

*More explicitly, any disjoint families of $p_1$ and $p_2$ blocks sensitive at zero for $f$ and $g$, respectively, yield a disjoint family of $p_1p_2$ blocks sensitive at zero for $f \circ g$.*

*Proof. Sensitivity.* For $x = (x_1, \dots, x_{n_1})$ put $y = (g(x_1), \dots, g(x_{n_1}))$. Flipping a coordinate inside $x_i$ leaves $y$ unchanged unless it changes $g(x_i)$, in which case $y$ becomes $y^{\{i\}}$. So the flip changes $f \circ g$ exactly when it is sensitive for $g$ at $x_i$ and $f(y^{\{i\}}) \neq f(y)$. Thus

$$s(f \circ g, x) = \sum_{i:\, f(y^{\{i\}}) \neq f(y)} s(g, x_i) \leq s(f, y) s(g) \leq s(f) s(g).$$

*Blocks.* Let $g(0) = 0$; then $(f \circ g)(0) = f(0, \dots, 0) = f(0)$. Let $X_1, \dots, X_{p_1} \subseteq [n_1]$ and $Y_1, \dots, Y_{p_2} \subseteq [n_2]$ be pairwise disjoint sensitive blocks at zero for $f$ and $g$ respectively. Identify the coordinates of $f \circ g$ with $[n_1] \times [n_2]$. The sets $X_i \times Y_j$ are nonempty and pairwise disjoint. Flipping $X_i \times Y_j$ at zero turns each child indexed by $X_i$ into $0^{Y_j}$, where $g = 1$ (as $g(0) = 0$ and $Y_j$ is sensitive), and leaves the other children at 0, where $g = 0$. The outer input becomes $0^{X_i}$, the indicator string of $X_i$, and $f(0^{X_i}) \neq f(0)$ because $X_i$ is sensitive for $f$ at zero. So each $X_i \times Y_j$ is sensitive for $f \circ g$ at zero. Taking maximum families gives the inequality (trivial if either maximum is zero). □

**Corollary 4.3.** *There exist a real number $\alpha > 2$ and nonconstant total Boolean functions $F_m$, for integers $m \geq 1$, such that*

$$\mathrm{bs}(F_m, 0) \geq s(F_m)^\alpha \qquad \textit{for every } m, \qquad \mathrm{bs}(F_m, 0) \longrightarrow \infty.$$

*Proof.* Fix $d = 9$ (so $L = 11$), an integer $M \geq 9^{10}$, and an admissible labelling; let $f$ be the function from the proof of Theorem 1.1. Put

$$A = 22M^{10}, \qquad B = M^2 k^9.$$

By (9) and (8), $s(f) \leq A$ and $\mathrm{bs}(f, 0) \geq B$; also $f(0) = 0$. Moreover

$$\frac{B}{A^2} = \frac{M^2 k^9}{484 M^{20}} = \frac{(k/M^2)^9}{484} \geq \frac{2^9}{484} = \frac{128}{121} > 1.$$

So $A > 1$, $B > A^2$, and $\alpha = \log B / \log A > 2$. This $\alpha$ depends on the chosen $M$; it is fixed once $M$ is.

Set $F_1 = f$ and $F_{m+1} = f \circ F_m$. We show by induction that $F_m$ is a total function on $n^m$ bits ($n$ as in (7)) with

$$F_m(0) = 0, \qquad s(F_m) \leq A^m, \qquad \mathrm{bs}(F_m, 0) \geq B^m.$$

For $m = 1$ this was just shown. Given the claim for $m$, Lemma 4.2 with inner function $F_m$ (allowed since $F_m(0) = 0$) gives $F_{m+1}(0) = f(0) = 0$, $s(F_{m+1}) \leq s(f) s(F_m) \leq A^{m+1}$ and $\mathrm{bs}(F_{m+1}, 0) \geq \mathrm{bs}(f, 0)\, \mathrm{bs}(F_m, 0) \geq B^{m+1}$.

Since $\mathrm{bs}(F_m, 0) \geq B^m \geq 1$, $F_m$ has a sensitive block and is nonconstant, and $\mathrm{bs}(F_m, 0) \to \infty$ because $B > 1$. Since $A^\alpha = B$,

$$s(F_m)^\alpha \leq A^{m\alpha} = B^m \leq \mathrm{bs}(F_m, 0).$$

The same estimates give

$$\frac{\mathrm{bs}(F_m)}{s(F_m)^2} \geq \left(\frac{B}{A^2}\right)^m \geq \left(\frac{128}{121}\right)^m.$$

□

## 5 Verification checklist

No step of the source was left unverified in this rewrite. A sceptical reader should check these load-bearing points first.

1. **(DC) and (SF).** Each clause uses each child at most once, so one flip changes at most one of its conditions (Section 2; "Three tools" in the proof of Proposition 3.1). All of Parts A–D rest on this.
2. **Directions of nesting.** $p_\gamma = 0 \Rightarrow p_Q = 0$ in C2; $p_Q = 1 \Rightarrow p_\gamma = 1$ in $(\star)$; and in Parts B and D the pair $\gamma' < \gamma$ consists of two *distinct* valid indices ($1 \le \gamma' < \gamma < Q$), as (J′) requires.
3. **Step C2.** The deduction $a(i,j) = m_j$ for every edge inside $\mathcal{T}$ matches exactly the pattern forbidden by Lemma 2.1; the arithmetic of that lemma is $\binom{16}{2} = 120$ and $k^{16} r^{-104} \le 3^{16} M^{-20} < 1$.
4. **Steps C3–C4.** Out-degree at most one inside $\mathcal{G}$ gives $|\mathcal{G}| \le 3$ with at most one sink; non-sinks cost $J_1'$ (two predicates change), the sink costs $S_1'$.
5. **Part D.** Every witness in $\mathcal{G}$ costs $J_1'$, with $p_{\gamma'}$ as second predicate, so $J_{\ell,0}$ has no $S_1'$ term. This absence is what keeps $v_\ell$ small in Lemma 4.1.
6. **Normalisation.** The exponents in Step 2 of Lemma 4.1 (divide by $M^{\ell+b}$), and the final inequality $(3^{\ell+1} - 3 - 2\ell)/3^{d+1} < 1$, which uses $M \ge 9^{d+1}$ and $\ell \le d$.
7. **Lemma 2.2.** Uniqueness of the nearby centre ($2d + 2 < 2L$) and the gap of two between distinct radii.
8. **Final count.** $s(f) \le \max\{MS_{d,0}, S_{d,1}\}$ for the OR, $\mathrm{bs}(f,0) \ge M^2 k^d$, and $k/M^2 > 2$ (proof of Theorem 1.1); $B/A^2 \ge 128/121$ and the use of $F_m(0) = 0$ in Corollary 4.3.

## References


[ABDK+21] Scott Aaronson, Shalev Ben-David, Robin Kothari, Shravas Rao, and Avishay Tal. Degree vs. approximate degree and quantum implications of Huang's sensitivity theorem. In *Proceedings of the 53rd Annual ACM SIGACT Symposium on Theory of Computing*, STOC '21, pages 1330–1342. ACM, 2021. doi:10.1145/3406325.3451047.

[AP14] Andris Ambainis and Krišjānis Prūsis. A tight lower bound on certificate complexity in terms of block sensitivity and sensitivity. Technical Report TR14-027, Electronic Colloquium on Computational Complexity, July 2014. Revision 1, July 31, 2014; revised report.

[AS11] Andris Ambainis and Xiaoming Sun. New separation between $s(f)$ and $bs(f)$. Technical Report TR11-116, Electronic Colloquium on Computational Complexity, 2011. arXiv:1108.3494v1.

[Hua19] Hao Huang. Induced subgraphs of hypercubes and a proof of the Sensitivity Conjecture. *Annals of Mathematics*, 190(3):949–955, 2019. doi:10.4007/annals.2019.190.3.6.

[Mei26] Alexander Meiburg. Block Sensitivity can exceed Spectral Sensitivity Squared, 2026. arXiv:2608.00851v1, 1 August 2026.

[Nis89] Noam Nisan. CREW PRAMs and decision trees. In *Proceedings of the Twenty-First Annual ACM Symposium on Theory of Computing*, STOC '89, pages 327–335. ACM, 1989. doi:10.1145/73007.73038.

[NS94] Noam Nisan and Mario Szegedy. On the degree of Boolean functions as real polynomials. *Computational Complexity*, 4(4):301–313, 1994. doi:10.1007/BF01263419.

[Rub95] David Rubinstein. Sensitivity vs. block sensitivity of Boolean functions. *Combinatorica*, 15(2):297–299, 1995. doi:10.1007/BF01200762.

[Tal13] Avishay Tal. Properties and applications of Boolean function composition. In *Proceedings of the 2013 ACM Conference on Innovations in Theoretical Computer Science*, ITCS '13, pages 441–454. ACM, 2013. doi:10.1145/2422436.2422485.

[Vir11] Madars Virza. Sensitivity versus block sensitivity of Boolean functions. *Information Processing Letters*, 111(9):433–435, 2011. doi:10.1016/j.ipl.2011.02.001.

[Wel22] Jake Wellens. Relationships between the number of inputs and other complexity measures of Boolean functions. *Discrete Analysis*, 2022. Paper No. 20, 21 pp.; arXiv:2005.00566v2.